

NAME: Prabhkirat Singh Marwah

SEC: B

BATCH: B4

ROLL NO: 47

ML LAB : Lab 2

In [3]:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn import metrics
```

In [5]:

```
df=pd.read_csv('USA_Housing.csv')
df.head()
```

Out[5]:

	Avg. Area Income	Avg. Area House Age	Avg. Area Number of Rooms	Avg. Area Number of Bedrooms	Area Population	Price	Address
0	79545.458574	5.682861	7.009188	4.09	23086.800503	1.059034e+06	208 Michael Ferry Apt. 674\nLaurabury, NE 3701...
1	79248.642455	6.002900	6.730821	3.09	40173.072174	1.505891e+06	188 Johnson Views Suite 079\nLake Kathleen, CA...
2	61287.067179	5.865890	8.512727	5.13	36882.159400	1.058988e+06	9127 Elizabeth Stravenue\nDanieltown, WI 06482...
3	63345.240046	7.188236	5.586729	3.26	34310.242831	1.260617e+06	USS Barnett\nFPO AP 44820
4	59982.197226	5.040555	7.839388	4.23	26354.109472	6.309435e+05	USNS Raymond\nFPO AE 09386

In [6]:

```
df.head()
```

Out[6]:

	Avg. Area Income	Avg. Area House Age	Avg. Area Number of Rooms	Avg. Area Number of Bedrooms	Area Population	Price	Address
0	79545.458574	5.682861	7.009188	4.09	23086.800503	1.059034e+06	208 Michael Ferry Apt. 674\nLaurabury, NE 3701...
1	79248.642455	6.002900	6.730821	3.09	40173.072174	1.505891e+06	188 Johnson Views Suite 079\nLake Kathleen, CA...
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4	59982.197226	5.040555	7.839388	4.23	26354.109472	6.309435e+05	USNS Raymond\nFPO AE 09386

In [7]:

```
if 'Address' in df.columns:
    df.drop(['Address'], axis=1, inplace=True)
```

In [8]: df.describe()

Out[8]:

	Avg. Area Income	Avg. Area House Age	Avg. Area Number of Rooms	Avg. Area Number of Bedrooms	Area Population	Price
count	5000.000000	5000.000000	5000.000000	5000.000000	5000.000000	5.000000e+03
mean	68583.108984	5.977222	6.987792	3.981330	36163.516039	1.232073e+06
std	10657.991214	0.991456	1.005833	1.234137	9925.650114	3.531176e+05
min	17796.631190	2.644304	3.236194	2.000000	172.610686	1.593866e+04
25%	61480.562388	5.322283	6.299250	3.140000	29403.928702	9.975771e+05
50%	68804.286404	5.970429	7.002902	4.050000	36199.406689	1.232669e+06
75%	75783.338666	6.650808	7.665871	4.490000	42861.290769	1.471210e+06
max	107701.748378	9.519088	10.759588	6.500000	69621.713378	2.469066e+06

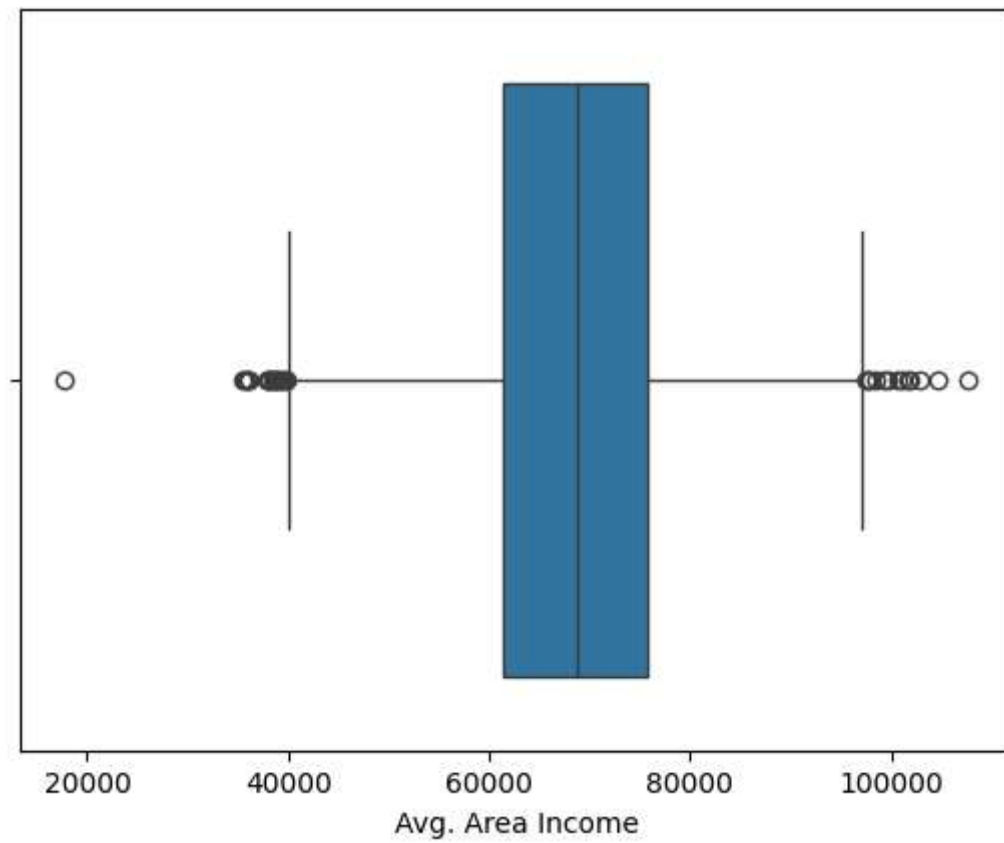
In [9]: pd.set_option('display.float_format', '{:2f}'.format)

In [10]: df.describe()

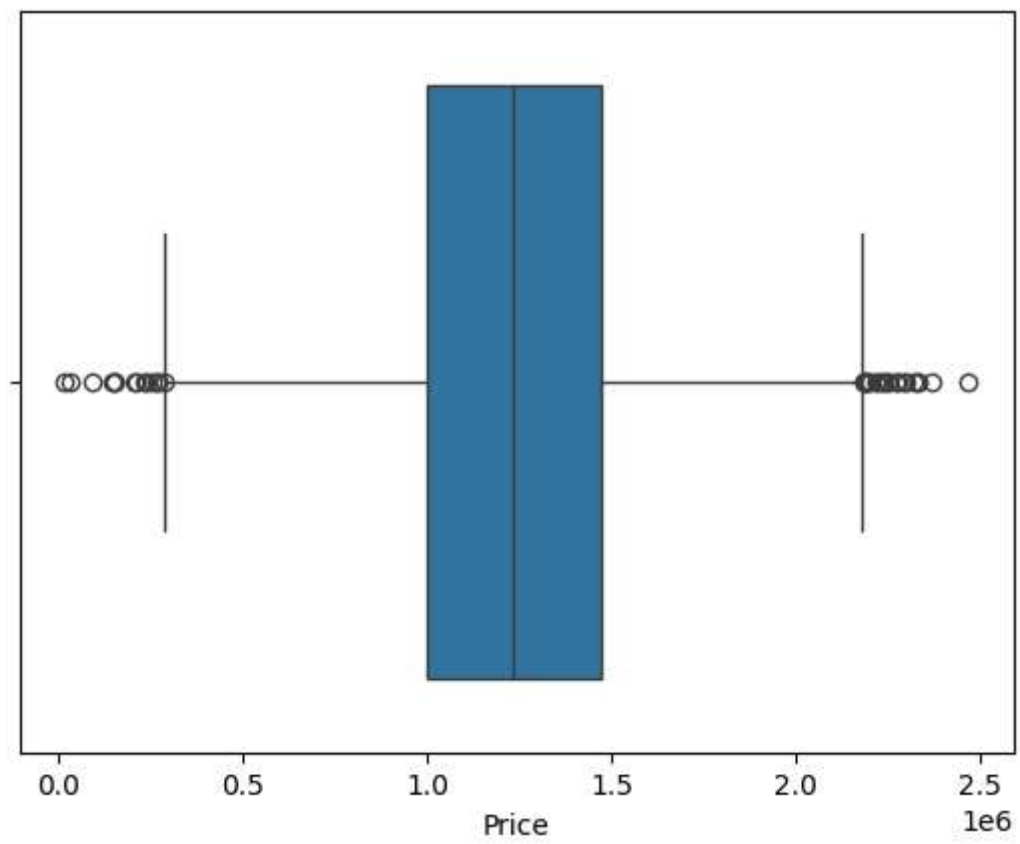
Out[10]:

	Avg. Area Income	Avg. Area House Age	Avg. Area Number of Rooms	Avg. Area Number of Bedrooms	Area Population	Price
count	5000.000000	5000.000000	5000.000000	5000.000000	5000.000000	5000.000000
mean	68583.108984	5.977222	6.987792	3.981330	36163.516039	1232072.654142
std	10657.991214	0.991456	1.005833	1.234137	9925.650114	353117.626581
min	17796.631190	2.644304	3.236194	2.000000	172.610686	15938.657923
25%	61480.562388	5.322283	6.299250	3.140000	29403.928702	997577.135049
50%	68804.286404	5.970429	7.002902	4.050000	36199.406689	1232669.377966
75%	75783.338666	6.650808	7.665871	4.490000	42861.290769	1471210.204212
max	107701.748378	9.519088	10.759588	6.500000	69621.713378	2469065.594175

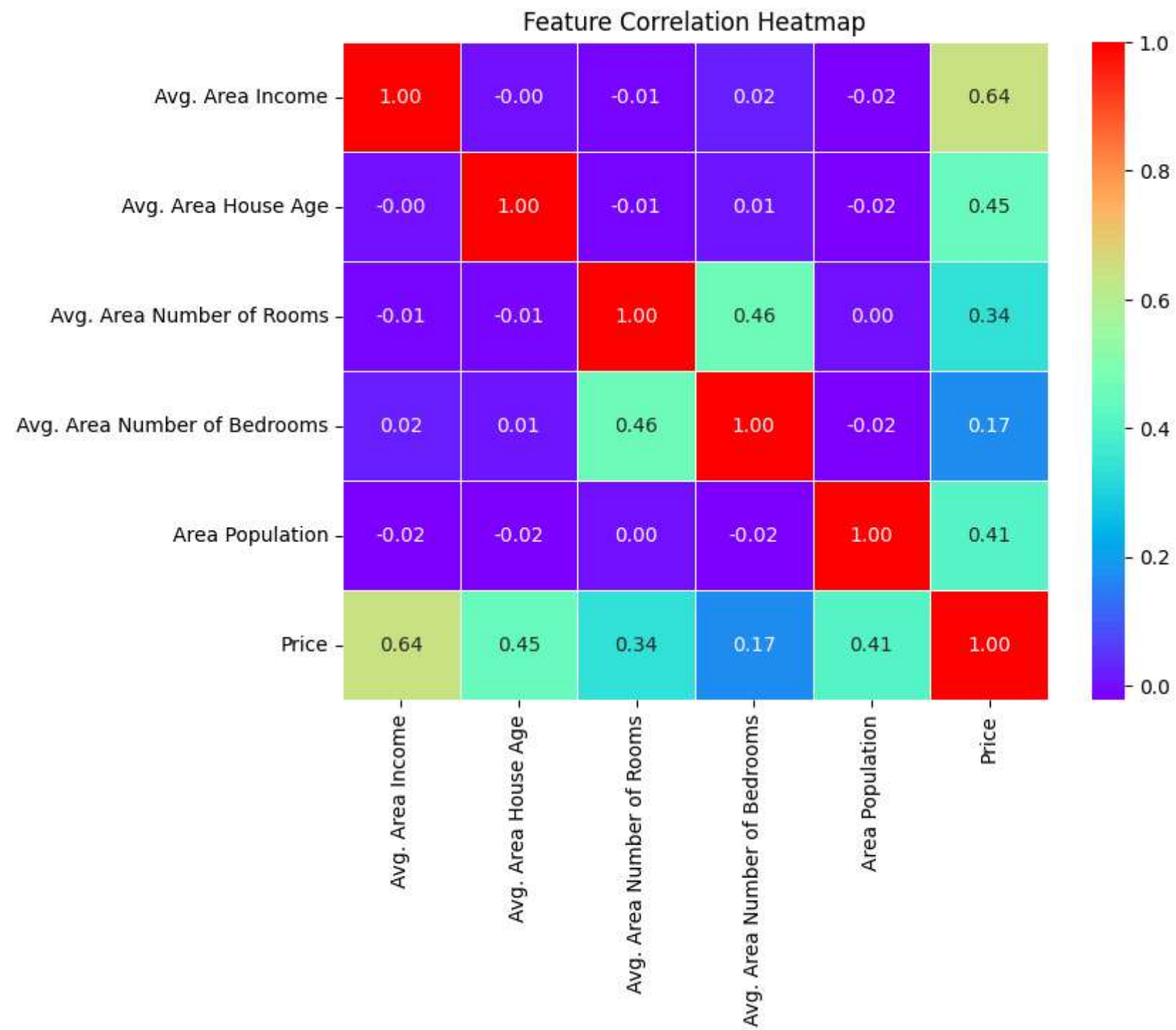
```
In [11]: ▶ sns.boxplot(x=df["Avg. Area Income"])
plt.show()
```



```
In [22]: ▶ sns.boxplot(x=df["Price"])
plt.show()
```



```
In [24]: ▶ plt.figure(figsize=(8,6))
corr_matrix = df.corr()
sns.heatmap(corr_matrix, cmap='rainbow',annot=True, fmt=".2f",linewidths=0.5)
plt.title("Feature Correlation Heatmap")
plt.show()
```



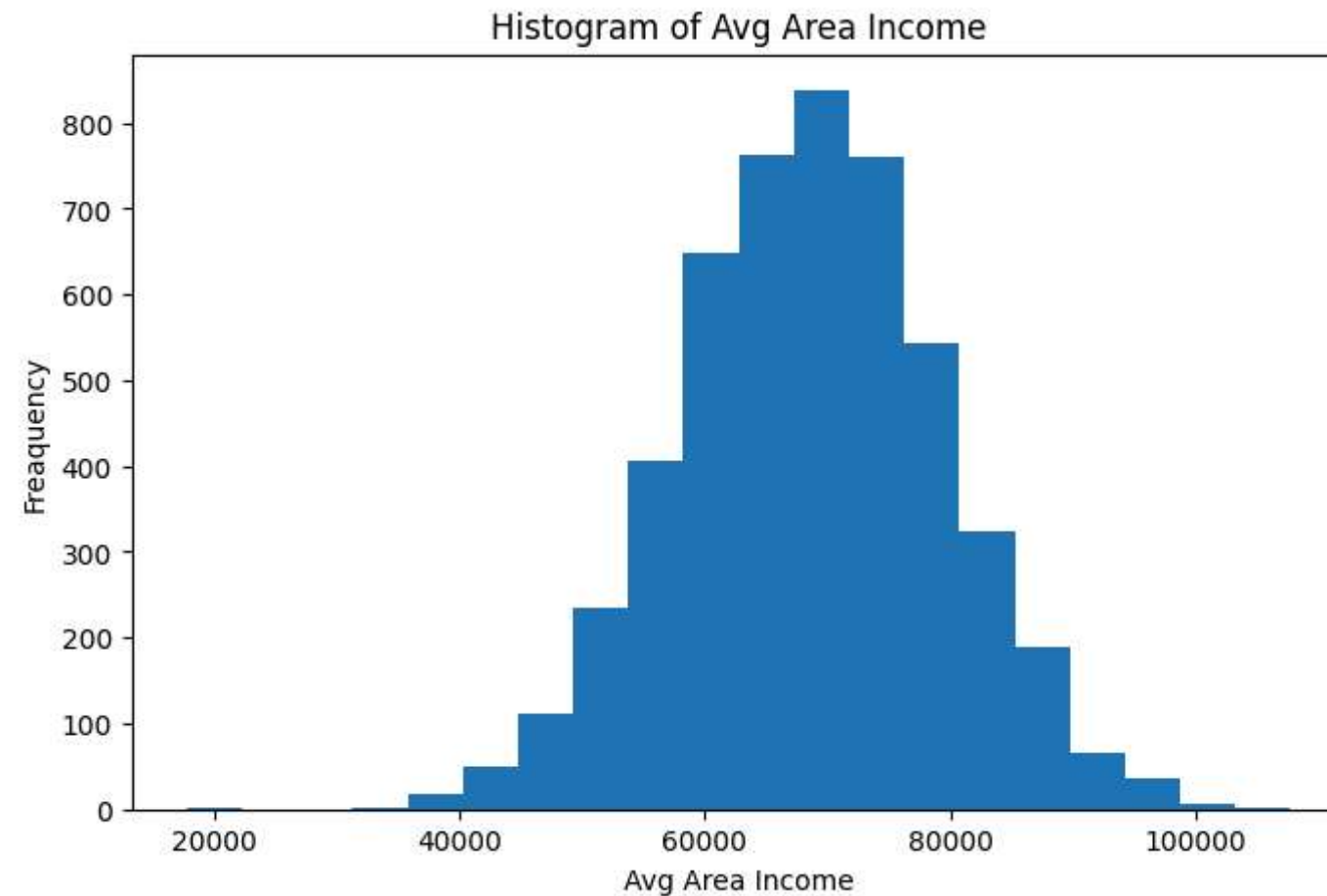
```
In [13]: sns.histplot(["Area population"],kde=True)
plt.show()
```



```
In [14]: df.isnull().sum()
```

```
Out[14]: Avg. Area Income      0
Avg. Area House Age      0
Avg. Area Number of Rooms  0
Avg. Area Number of Bedrooms  0
Area Population      0
Price      0
dtype: int64
```

```
In [15]: ▶ plt.figure(figsize=(8,5))
plt.hist(df['Avg. Area Income'], bins=20)
plt.title('Histogram of Avg Area Income')
plt.xlabel('Avg Area Income')
plt.ylabel('Freaquency')
plt.show()
```



SLR

```
In [25]: ▶ x = df[['Avg. Area Income']]
y = df['Price']
```

```
In [26]: ▶ from sklearn.model_selection import train_test_split
x_train , x_test, y_train, y_test = train_test_split(x,y,test_size=0.2,random_state=42)
```

```
In [27]: ▶ from sklearn.linear_model import LinearRegression
slr_model = LinearRegression()
slr_model.fit(x_train,y_train)
```

Out[27]: LinearRegression()

**In a Jupyter environment, please rerun this cell to show the HTML representation or trust the notebook.
On GitHub, the HTML representation is unable to render, please try loading this page with nbviewer.org.**

```
In [28]: ▶ y_pred = slr_model.predict(x_test)
```



```
In [31]: ► from sklearn.metrics import mean_absolute_error
from sklearn.metrics import mean_squared_error
from sklearn.metrics import r2_score
from sklearn.metrics import root_mean_squared_error
import numpy as np

print('simple linear regression results')
print('-----')
```

```
simple linear regression results
-----
```

```
In [32]: ► print("MAE: ", mean_absolute_error(y_test, y_pred))
print("MSE: ", mean_squared_error(y_test, y_pred))
print("RMSE: ", root_mean_squared_error(y_test, y_pred))
print("R2 Score: ", r2_score(y_test, y_pred))
```

```
MAE:  216826.35309989064
MSE:  74194887058.66
RMSE:  272387.38417676394
R2 Score:  0.39694865192619766
```

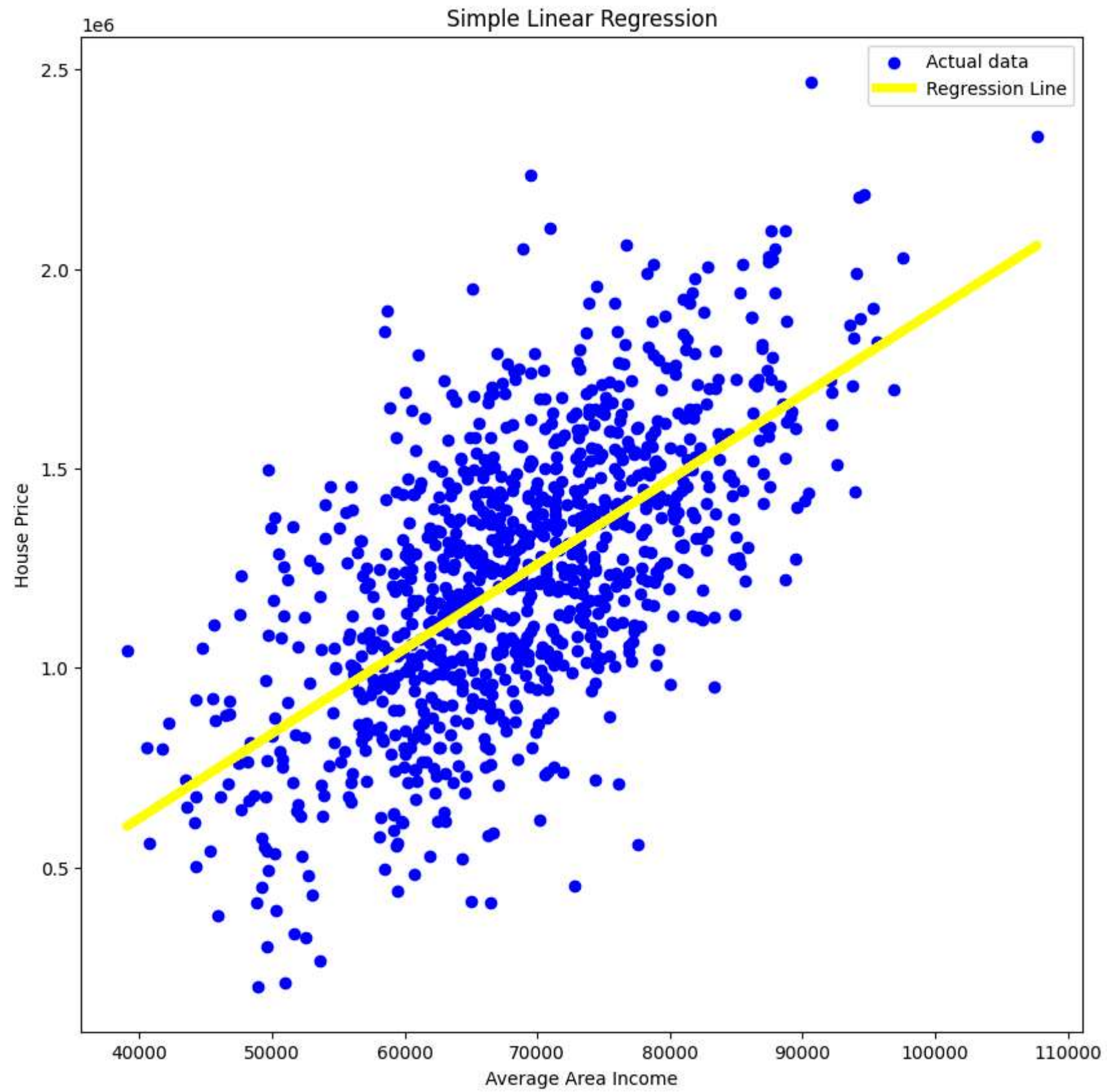
```
In [34]: ► print('intercept: ', slr_model.intercept_)
print('slope:', slr_model.coef_[0])
```

```
intercept:  -225053.70287170145
slope:  21.208906292657513
```

```
In [46]: ▶ plt.figure(figsize=(8,6))

plt.figure(figsize=(10,10))
plt.scatter(x_test,y_test, color='blue',label='Actual data')
plt.plot(x_test, y_pred,color='yellow', linewidth=5,label='Regression Line')
plt.xlabel("Average Area Income")
plt.ylabel("House Price")
plt.title("Simple Linear Regression")
plt.legend()
plt.show()
```

<Figure size 800x600 with 0 Axes>



```
In [41]: income = float(input("Enter Avg ARea Income"))
new_data = np.array([[income]])
prediction = slr_model.predict(new_data)
print("Predicted House Price =${:,.2f}".format(prediction[0]))
```

Enter Avg ARea Income 13212

Predicted House Price =\$55,158.37

C:\Users\prabh\AppData\Roaming\Python\Python313\site-packages\sklearn\utils\validation.py:2827: UserWarning: X does not have valid feature names, but LinearRegression was fitted with feature names
warnings.warn(

```
In [ ]:
```